

Intepreting 'one-unit' Change in Logistic Regression [duplicate]

Asked 8 years, 5 months ago Modified 5 years, 4 months ago Viewed 6k times

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[Interpretation of simple predictions to odds ratios in logistic regression](#) (2 answers)

Closed 8 years ago.

I have very easy question that I'm hoping someone can assist me with:

I ran an example logistic regression using this R code:

```
hours <- c(0.5, 0.75, 1, 1.25, 1.5, 1.75, 1.75, 2, 2.25, 2.5, 2.75, 3, 3.25,
3.5, 4, 4.25, 4.5, 4.75, 5, 5.5)
pass <- c(0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1)
data <- data.frame(hours, pass)
mylogit <- glm(pass ~ hours, data = data, family = "binomial") #Activates
the logistic regression model
summary(mylogit) #Summary of the model

Call:
glm(formula = pass ~ hours, family = "binomial", data = data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.70557  -0.57357  -0.04654   0.45470   1.82008

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -4.0777      1.7610  -2.316  0.0206 *
hours         1.5046      0.6287   2.393  0.0167 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 27.726  on 19  degrees of freedom
Residual deviance: 16.060  on 18  degrees of freedom
AIC: 20.06

Number of Fisher Scoring iterations: 5

round(exp(cbind(OR = coef(mylogit), confint(mylogit))),3)

              OR 2.5 % 97.5 %
(Intercept) 0.017 0.000 0.281
hours       4.503 1.698 23.223
```

I know that by taking the exponent of the log-odds/coefficients for hours the odds of passing increase by a factor of 4.503 for a one-unit change in hours. However, given that the explanatory variable (hours) is continuous, what is considered a 'one-unit change' i.e. going from 1 to 2 hours as one unit? or from 1.75 to 1.76 hours as one unit? Also, is this interpretation of one-unit the same for regular OLS regression as well? I'm seeking to better understand the rules R applies to creating its regression coefficients.

r regression logistic interpretation odds-ratio

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edited Apr 11, 2019 at 18:34

 **Krantz**
585 5 19

asked Mar 4, 2016 at 22:26

 **DaBenski**
57 1 5

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- The change is one-unit on the same scale as your explanatory variable. Thus a one-unit change in your case is 1 hour. The effect is the same all along the scale (ie. from 1 to 2 hours or from 4 to 5 hours) – [ekstroem](#) Mar 4, 2016 at 22:33

1 Answer

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With logistic regression you analyse the association of a binary outcome with a set of predictors. In particular you are modelling the $\log(\text{odds})$ of that particular outcome. The odds themselves are simply: $\text{odds} = \frac{\text{Pr}(\text{of Occurring})}{\text{Pr}(\text{of Not Occurring})} = \frac{\text{Pr}(\text{of Occurring})}{1 - \text{Pr}(\text{of Occurring})}$ where Pr refers to proportions (or loosely speaking probability). Therefore the estimated coefficient $\hat{\beta}$ in your model shows the difference in $\log(\text{odds})$ between two subjects that differ by one unit of your predictor (here: `hours`) when all other predictors are constant (or as in your case just absent).

To find the change in terms of the proportions that are modelled you need to:

1. Get the $\log(\text{odds})$ estimate.
2. Exponentiate it to get the odds.
3. Get the new proportions as: $\text{Pr}_{\text{new}} = \frac{\text{odds}}{1 + \text{odds}}$. (This follows from the equality shown above.)

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edited Mar 5, 2016 at 0:43

 **Nick Cox**
58.4k 8 134 199

answered Mar 4, 2016 at 23:16

 **user11852**
45.1k 3 103 162

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- Thanks for the detailed answer about interpreting the probability. However, what I am really trying to understand is the definition of a 'one unit' change in terms of a continuous predictor variable. Is it fractional? or an integer as [@ekstroem](#) suggested? – [DaBenski](#) Mar 7, 2016 at 16:18

1  When referring to "*one unit of change*" in the predictor one refers at adding or subtracting 1 from a continuous predictor variable in the scale that this variable is recorded (eg. `hours` in this case). Notice that the change to the predicted proportions is not quantized in the same way as the `log(odd)` transform is not linear. – [user11852](#) Mar 7, 2016 at 17:38

 Makes sense now - @user11852 I appreciate your help. – [DaBenski](#) Mar 7, 2016 at 17:50